NUS-ISS Practice Module Brief: Intelligent Reasoning Systems

Brief · Coursework · NUS-ISS teaching team · 2026-07-06

Course brief for the Intelligent Reasoning Systems practice module: a group-built reasoning system with a report, a runnable demo and a short video.

Purpose

The practice module asks each group to design and build an intelligent reasoning system that solves a real problem for a real user.

Systems should combine at least two reasoning techniques, such as knowledge representation, search, machine learning or retrieval, and explain how they work together.

Deliverables

Each group submits a project report, a runnable demo web app and a recorded video of no more than ten minutes.

The runnable demo must start with documented commands on a clean machine, and markers must be able to explore it without private credentials.

Provenance matters: every answer a system gives should be traceable back to the evidence it used.

Assessment

Marks are awarded for problem framing, technical depth, the quality of the reasoning, evaluation against ground truth, and the clarity of the demo.

Groups are encouraged to keep architecture decisions in a written log and to show how evaluation shaped the design.